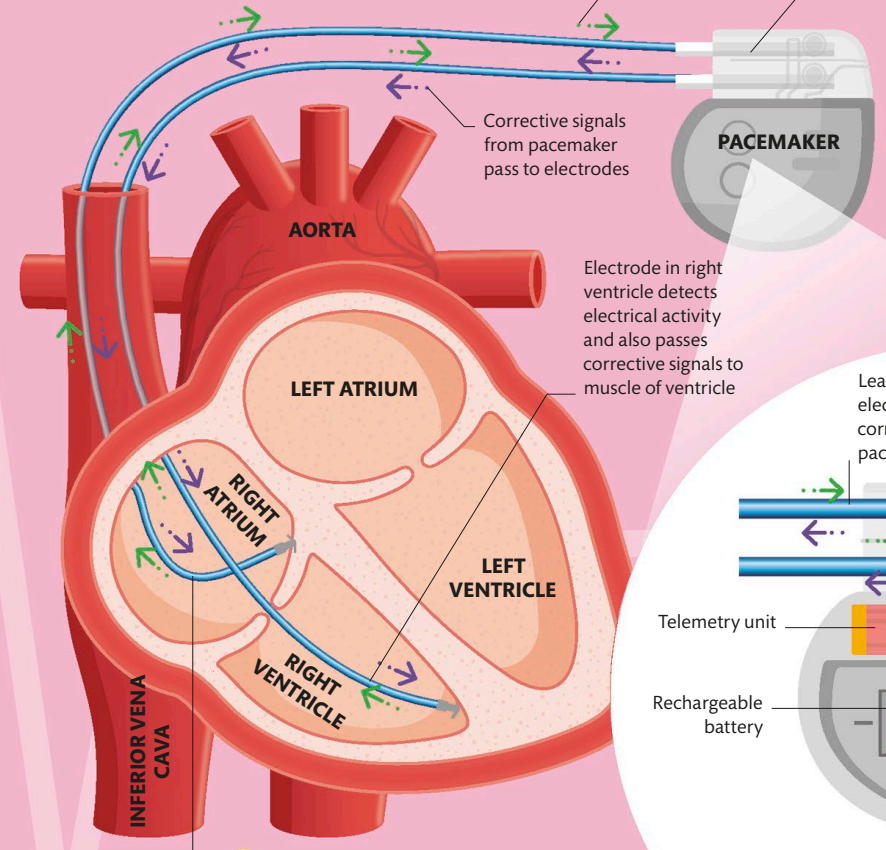
WORLDWIDE, MORE THAN 1 MILLION PACEMAKERS ARE IMPLANTED EVERY YEAR

ICD

An implantable cardioverter defibrillator (ICD) is fitted to people at risk of life-threatening abnormal heart rhythms. Like a pacemaker, an ICD can detect very fast or chaotic heartbeats; in such cases, the ICD gives the heart a small electric shock (cardioversion) or a larger shock (defibrillation) to reestablish a normal heart rhythm. Sometimes, an ICD may be combined with a pacemaker.

Dual-chamber pacemaker
This device has two leads: one for the right atrium and one for the right ventricle. It is used to correct faulty signals from the heart's nodes that cause an abnormal heartbeat rhythm. By sending out corrective signals, the pacemaker makes the heart's chambers contract in a normal rhythm.

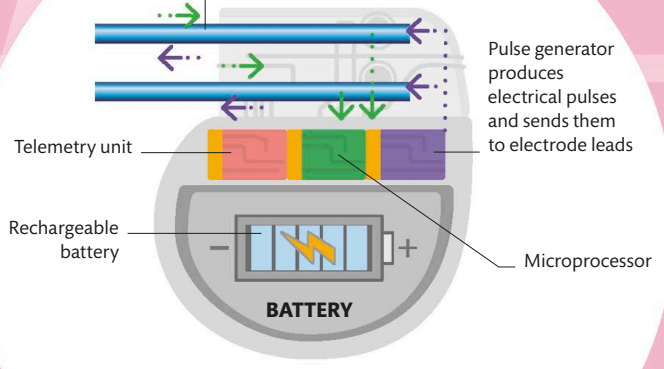
1 Pacemaker monitors heart
Electrodes inside the heart's chambers constantly monitor electrical signals in the heart and send data about this activity to a microprocessor inside the pacemaker. The microprocessor is programmed to recognize when signals are abnormal or missing.



2 Pacemaker detects abnormal activity
When the microprocessor identifies abnormal signals, it instructs the pulse generator in the pacemaker to transmit low-voltage electrical pulses to the electrodes in the heart. The pulses stimulate the muscle in the heart's chambers to contract.

3 Abnormal activity corrected
Once the heartbeat has returned to normal, the pacemaker stops sending electrical pulses. However, it continues to monitor the heart and collect data. This information can be relayed to an external computer, enabling doctors to assess how well the pacemaker is working.

INSIDE A PACEMAKER



A microprocessor regulates electrical pulses sent out by a pulse generator and also contains a memory and a monitor to collect data about the heart's activity. Connected to the microprocessor is a telemetry unit, which exchanges data with an external computer. Power is provided by a rechargeable battery.